Predicting Pathologic Complete Response from Clinical Variables

Tom Stoughton, MS Candidate[1]

Abstract

...

Keywords: Pathologic Complete Response, Breast Cancer, Logistic Regression,

Introduction

This data set contains publicly available clinical and demographic information about 1,506 breast cancer patients from four different trials[1]

Motivating Problem

...

Methods

Using Multivariate Imputation by Chained Equations (MICE).

logistic regression on clinical variables.

Table 1: Model 1: Pooled Logistic Regression

Variable	estimate	std.error	statistic	df	P_Value
(Intercept)	**-1.574**	**0.275**	**-5.719**	**1335.758**	**0.000**
multifocal_cancer	-0.109	0.132	-0.832	476.424	0.406
bilateral_breast_cancer	-1.026	0.557	-1.842	1490.747	0.066
bmi_groupobesity_class_1	0.011	0.180	0.061	857.202	0.951
bmi_groupobesity_class_2	0.129	0.263	0.490	290.248	0.625
bmi_groupobesity_class_3	-0.117	0.292	-0.401	551.110	0.689
bmi_groupoverweight	**0.461**	**0.154**	**2.992**	**648.354**	**0.003**
bmi_groupunderweight	**1.233**	**0.376**	**3.275**	**362.539**	**0.001**
her2	**1.237**	**0.206**	**6.000**	**1454.648**	**0.000**
tumor_subtypeluminal	-0.405	0.215	-1.880	1432.443	0.060
tumor_subtypetriple_negative	**0.837**	**0.264**	**3.163**	**1324.640**	**0.002**
menopausepre	0.121	0.129	0.937	590.632	0.349

Now for including demographics (age and ethnicity).

[1]Department of Biostatistics and Informatics, Colorado School of Public Health, University of Colorado Anschutz Medical Campus, Aurora, CO

Table 2: Model 2: Pooled Logistic Regression (with Demographics)

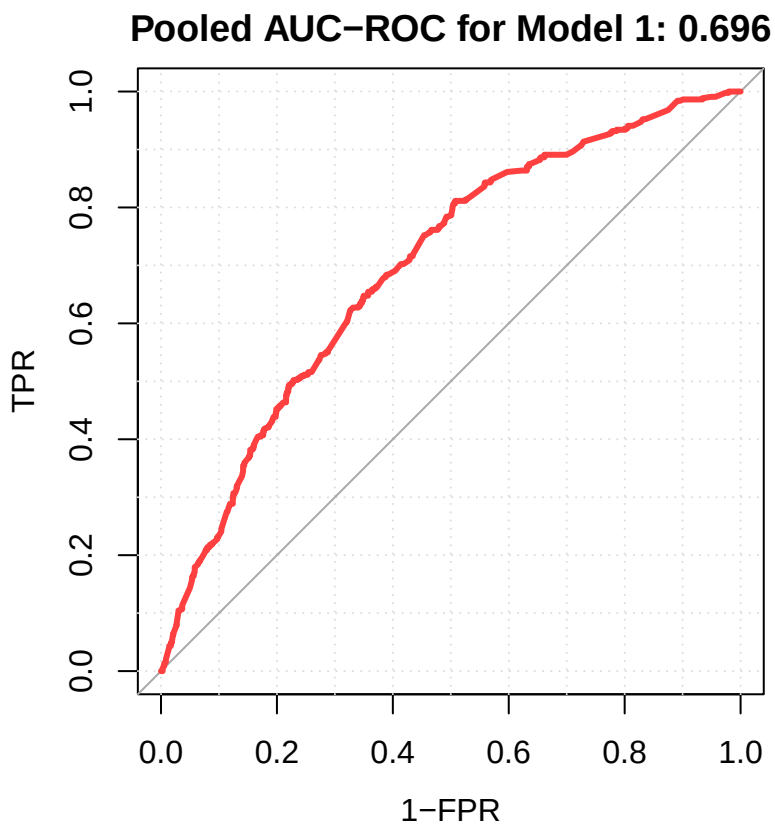
Variable	estimate	std.error	statistic	df	P_Value
(Intercept)	-1.089	0.601	-1.813	617.600	0.070
multifocal_cancer	-0.113	0.133	-0.850	444.089	0.396
bilateral_breast_cancer	-1.086	0.560	-1.937	1484.602	0.053
bmi_groupobesity_class_1	0.056	0.184	0.303	1004.906	0.762
bmi_groupobesity_class_2	0.198	0.270	0.733	268.015	0.464
bmi_groupobesity_class_3	-0.030	0.301	-0.100	545.358	0.920
bmi_groupoverweight	**0.492**	**0.156**	**3.160**	**662.387**	**0.002**
bmi_groupunderweight	**1.294**	**0.378**	**3.425**	**396.908**	**0.001**
age	-0.012	0.009	-1.334	373.728	0.183
her2	**1.218**	**0.208**	**5.861**	**1431.585**	**0.000**
tumor_subtypeluminal	**-0.475**	**0.219**	**-2.171**	**1374.021**	**0.030**
tumor_subtypetriple_negative	**0.782**	**0.268**	**2.921**	**1240.794**	**0.004**
menopausepre	-0.060	0.204	-0.295	189.513	0.768
ethnicityamerican indian/alaska native	-0.588	1.105	-0.532	1440.171	0.595
ethnicityasian	0.222	0.305	0.729	1339.217	0.466
ethnicitycaucasian	0.298	0.179	1.660	1111.091	0.097
ethnicityhawaiian/pacific islander	**1.980**	**0.811**	**2.440**	**1466.871**	**0.015**
ethnicityhispanic	-0.921	0.861	-1.070	866.769	0.285

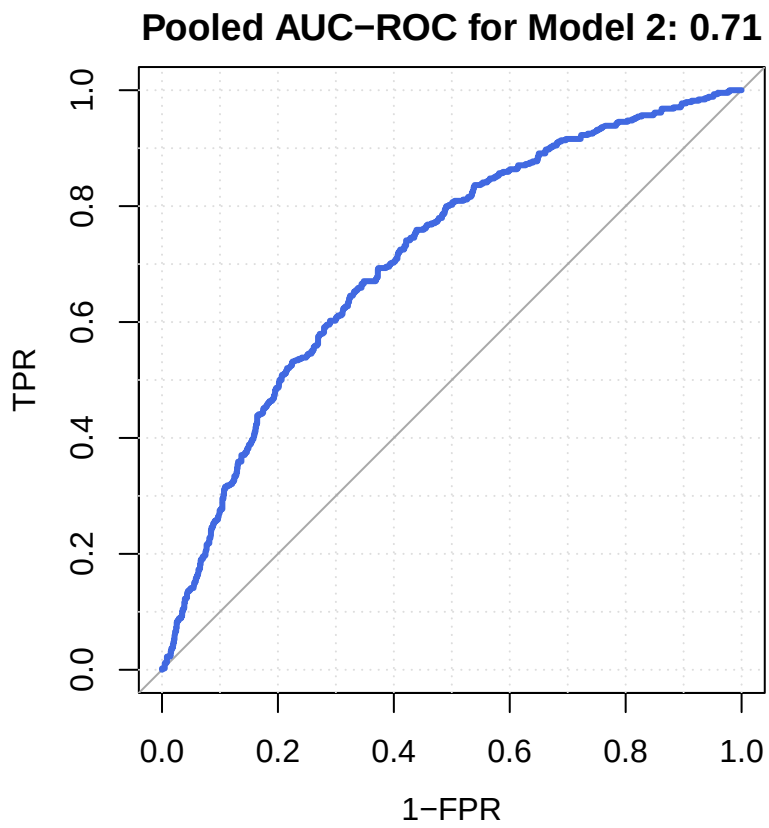
Now for including treatment agents.

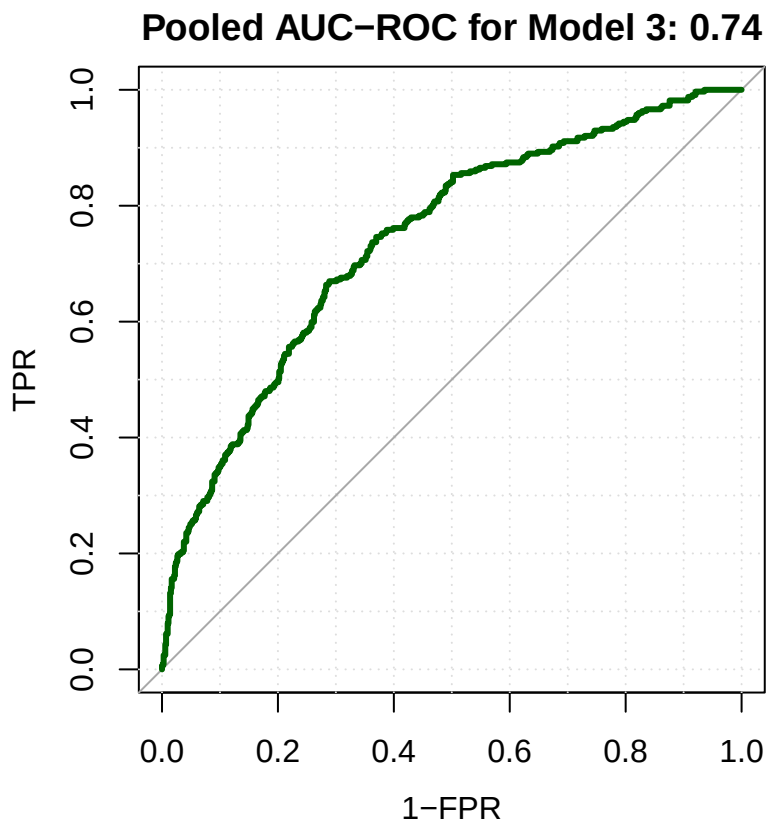
Table 3: Model 3: Pooled Logistic Regression (Clinical + Treatment Agents)

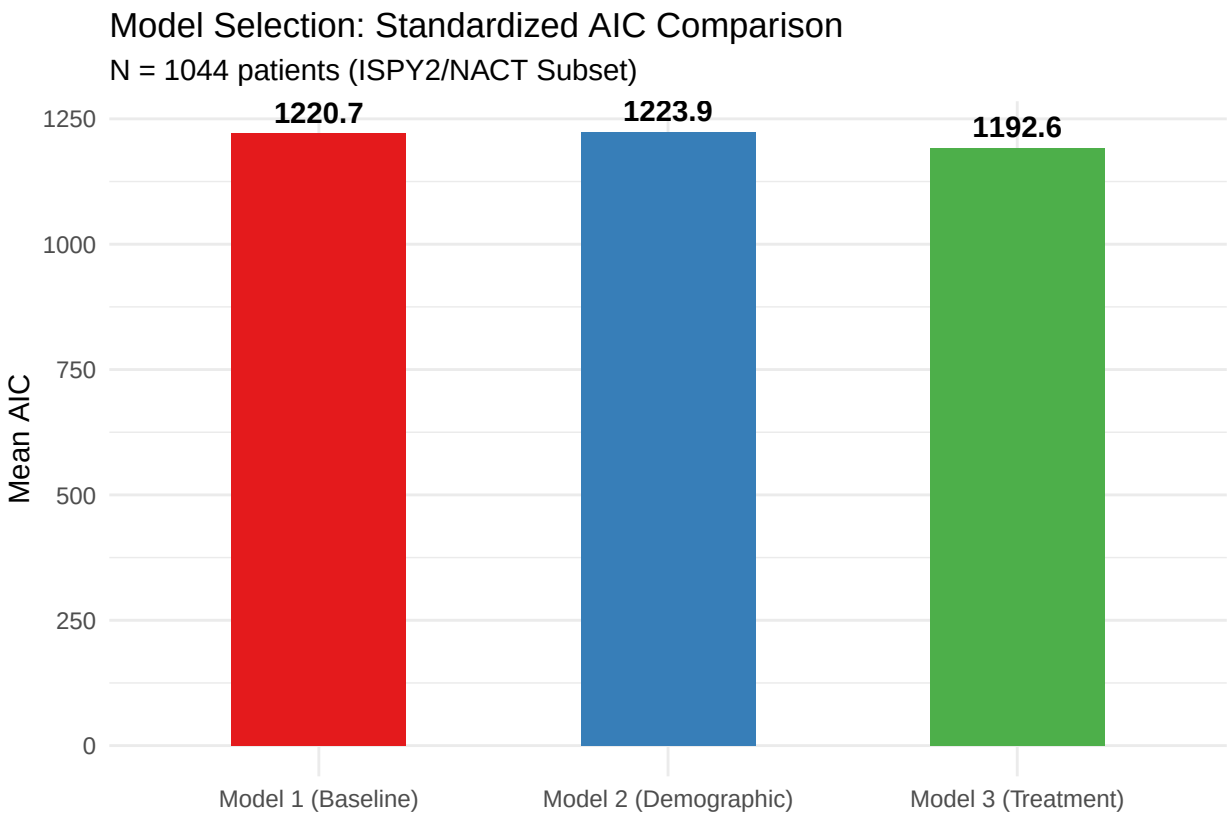
Variable	estimate	std.error	statistic	df	P_Value
(Intercept)	-14.365	1005.594	-0.014	1010.006	0.989
bilateral_breast_cancer	-0.881	0.654	-1.346	1005.164	0.179
multifocal_cancer	-0.272	0.158	-1.719	994.444	0.086
her2	0.487	0.425	1.144	911.573	0.253
tumor_subtypeluminal	**-1.202**	**0.294**	**-4.088**	**976.305**	**0.000**
tumor_subtypetriple_negative	0.004	0.347	0.010	958.755	0.992
age	-0.018	0.012	-1.528	251.025	0.128
bmi_groupobesity_class_1	0.023	0.224	0.101	544.808	0.919
bmi_groupobesity_class_2	0.085	0.344	0.247	172.273	0.806
bmi_groupobesity_class_3	0.018	0.394	0.045	553.430	0.964
bmi_groupoverweight	**0.391**	**0.197**	**1.988**	**279.465**	**0.048**
bmi_groupunderweight	**1.576**	**0.504**	**3.127**	**299.760**	**0.002**
menopausepre	-0.252	0.262	-0.962	147.460	0.337
ethnicityamerican indian/alaska native	-1.421	1.242	-1.144	1008.889	0.253
ethnicityasian	-0.025	0.367	-0.067	964.967	0.946
ethnicitycaucasian	0.102	0.243	0.418	882.084	0.676
ethnicityhawaiian/pacific islander	0.854	0.991	0.862	987.896	0.389
ethnicityOther/Unknown	-14.722	756.652	-0.019	1010.006	0.984
agent_paclitaxel	14.168	1005.594	0.014	1010.005	0.989
agent_ganitumab	0.432	0.310	1.394	1006.380	0.163
agent_ganetespib	**0.695**	**0.312**	**2.224**	**1004.136**	**0.026**
agent_mk_2206	**0.870**	**0.302**	**2.877**	**993.059**	**0.004**
agent_pembrolizumab	**1.679**	**0.330**	**5.096**	**998.047**	**0.000**
agent_amg_386	**0.670**	**0.275**	**2.436**	**998.777**	**0.015**
agent_trastuzumab	0.337	0.477	0.707	830.615	0.480
agent_neratinib	**0.797**	**0.359**	**2.217**	**942.059**	**0.027**
agent_t_dm1	14.420	1005.594	0.014	1010.006	0.989
agent_pertuzumab	**1.531**	**0.443**	**3.459**	**1000.229**	**0.001**
agent_anthracycline	13.861	1005.594	0.014	1010.006	0.989
agent_taxane	0.402	0.779	0.516	839.804	0.606
*agent_taxane___incomplete_*	-15.610	860.703	-0.018	1010.006	0.986
agent_abt_888_carboplatin	**1.071**	**0.328**	**3.264**	**1001.530**	**0.001**

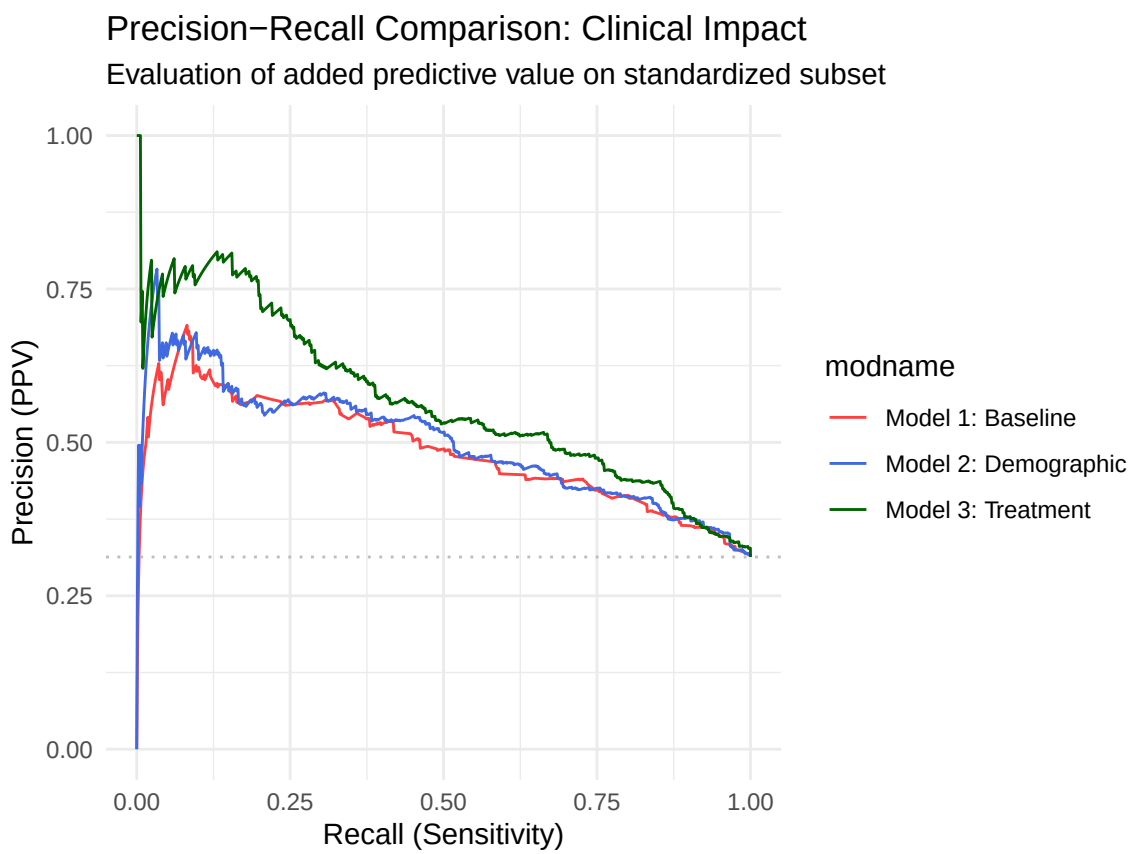
Results

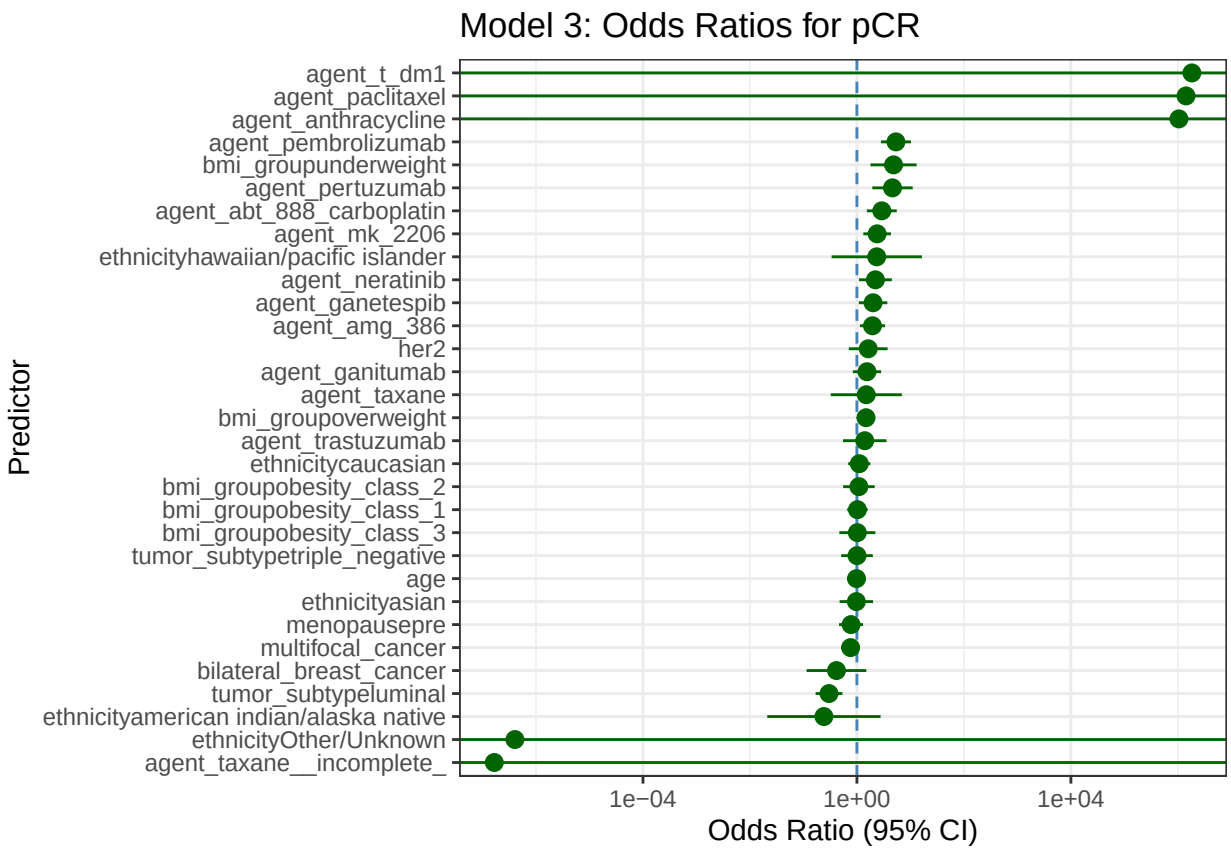



Pooled AUC−ROC for Model 2: 0.71
TPR
1−FPR
0.0
0.2
0.4
0.6
0.8
1.0


Pooled AUC−ROC for Model 3: 0.74
TPR
1−FPR
0.0
0.2
0.4
0.6
0.8
1.0


Model Selection: Standardized AIC Comparison
N = 1044 patients (ISPY2/NACT Subset)
Mean AIC
1250
1000
750
500
250
0
1220.7
1223.9
1192.6
Model 1 (Baseline)
Model 2 (Demographic)
Model 3 (Treatment)


Precision−Recall Comparison: Clinical Impact
Evaluation of added predictive value on standardized subset
Precision (PPV)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Recall (Sensitivity)
modname
Model 1: Baseline
Model 2: Demographic
Model 3: Treatment


Model 3: Odds Ratios for pCR
agent_t_dm1
agent_paclitaxel
agent_anthracycline
agent_pembrolizumab
bmi_groupunderweight
agent_pertuzumab
agent_abt_888_carboplatin
agent_mk_2206
ethnicityhawaiian/pacific islander
agent_neratinib
agent_ganetespib
agent_amg_386
her2
agent_ganitumab
agent_taxane
bmi_groupoverweight
agent_trastuzumab
ethnicitycaucasian
bmi_groupobesity_class_2
bmi_groupobesity_class_1
bmi_groupobesity_class_3
tumor_subtypetriple_negative
age
ethnicityasian
menopausepre
multifocal_cancer
bilateral_breast_cancer
tumor_subtypeluminal
ethnicityamerican indian/alaska native
ethnicityOther/Unknown
agent_taxane__incomplete_
Predictor
1e−04
1e+00
1e+04
Odds Ratio (95% CI)

Conclusion

...

References

1. Garrucho L, Kushibar K, Reidel C-A, Joshi S, Osuala R, Tsirikoglou A, et al. A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations. Scientific Data 2025;12:453. https://doi.org/10.1038/s41597-025-04707-4.

Code Appendix

```
knitr::opts_chunk$set(echo = F)

library(tidyverse)
library(dplyr)
library(recipes)
library(kableExtra)
library(gtsummary)
library(ggplot2)
library(pROC)
library(precrec)
library(mice)

clinicalData <- read.csv("clinical_and_imaging_info.csv") %>%
  mutate(
    pcr = replace_na(pcr, replace=0)
  )
relevantData <- clinicalData %>%
  select(
    pcr, patient_id,
    dataset, nac_agent,
    bilateral_breast_cancer,
    multifocal_cancer,
    hr, er, pr, her2,
    nottingham_grade,
    tumor_subtype,
    age, weight, bmi_group,
    menopause, ethnicity
  )
# clean missing data
relevantData <- relevantData %>%
  mutate(across(where(is.character), ~na_if(str_trim(.), "")))

# clean up data
relevantData <- relevantData %>% select(
  patient_id, pcr, dataset, nac_agent,
  bilateral_breast_cancer,
  multifocal_cancer,
  her2, tumor_subtype,
  age, bmi_group,
  menopause, ethnicity
) %>% mutate(
  menopause = case_when(
    str_detect(menopause, "(?i)pre|peri") ~ "pre",
    str_detect(menopause, "(?i)post")     ~ "post",
```

```r
    is.na(menopause)                      ~ NA_character_,
  ),
  tumor_subtype = case_when(
    str_detect(tumor_subtype, "(?i)luminal") ~ "luminal",
    str_detect(tumor_subtype, "(?i)her2")    ~ "her2_enriched",
    is.na(tumor_subtype)                     ~ NA_character_,
    TRUE                                     ~ tumor_subtype
  ),

  ethnicity = case_when(
    str_detect(ethnicity, "(?i)american indian|native") ~ "american indian/alaska native",
    str_detect(ethnicity, "(?i)hawai|pacific islander") ~ "hawaiian/pacific islander",
    str_detect(ethnicity, "(?i)multiple race")          ~ NA_character_,
    is.na(ethnicity)                                    ~ NA_character_,
    TRUE                                                ~ ethnicity
  )
)

# Now one-hot encode treatment variables
unique_agents <- relevantData %>%
  filter(!is.na(nac_agent) & nac_agent != "") %>%
  pull(nac_agent) %>%
  str_split(" \\+ ") %>%  # Splits by " + "
  unlist() %>%
  str_trim() %>%
  unique()

for (agent in unique_agents) {
  # Create a clean column name
  col_name <- paste0("agent_", str_replace_all(tolower(agent), "[^a-z0-9]", "_"))

  relevantData <- relevantData %>%
    mutate(!!col_name := case_when(
      is.na(nac_agent) ~ NA_real_,
      stringr::str_detect(as.character(nac_agent), stringr::fixed(agent)) ~ 1,
      TRUE ~ 0
    ))
}

relevantData <- relevantData %>% select(-nac_agent)
write.csv(relevantData, "./processedData.csv", row.names=F)
data <- read.csv("./processedData.csv")
dataNoMissing <- data %>% drop_na()

# Impute the missing data (m=5 creates 5 versions of the data)
data_for_imputation <- data %>%
  mutate(across(where(is.character), as.factor)) %>%
  select(-starts_with("agent_"))

# 2. Initialize the setup without running the iterations yet
ini <- mice(data_for_imputation, maxit = 0)

# 3. Modify the Predictor Matrix
# Setting a column to 0 in the predictorMatrix means it won't be used to predict anything else
```

```r
pred <- ini$predictorMatrix
pred[, "patient_id"] <- 0  # patient_id will not be used as a predictor
pred["patient_id", ] <- 0  # patient_id itself will not be imputed

# 4. Modify the Method
# Setting the method to "" tells mice to ignore this column during the math
meth <- ini$method
meth["patient_id"] <- ""

# 5. Run the actual imputation with these settings
imputed_data <- mice(data_for_imputation,
                     m = 10,
                     method = meth,
                     predictorMatrix = pred,
                     seed = 123)

# Run your logistic regression on all 5 datasets and pool the results
fit1 <- with(imputed_data,
             glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                 bmi_group + her2 + tumor_subtype + menopause,
                 family = binomial))
pooled_results1 <- pool(fit1)

reg_table_data <- summary(pooled_results1) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(Variable = term, P_Value = p.value)

# 2. Identify rows where p-value < 0.05
sig_rows <- which(reg_table_data$P_Value < 0.05)

# 3. Create the table
reg_table_data %>%
  kbl(caption = "Model 1: Pooled Logistic Regression",
      booktabs = TRUE) %>%
  kable_styling(bootstrap_options = c("striped", "hover"),
                full_width = FALSE,
                latex_options = "HOLD_position") %>%
  row_spec(sig_rows, bold = TRUE) %>% # Bolds significant predictors like Pembrolizumab
  column_spec(1, italic = TRUE)

# Run your logistic regression on all 5 datasets and pool the results
fit2 <- with(imputed_data,
             glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                  bmi_group + age + her2 + tumor_subtype +
                  menopause + ethnicity, family = binomial))
pooled_results2 <- pool(fit2)

reg_table_data <- summary(pooled_results2) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(Variable = term, P_Value = p.value)

# 2. Identify rows where p-value < 0.05
sig_rows <- which(reg_table_data$P_Value < 0.05)
```

```r
# 3. Create the table
reg_table_data %>%
  kbl(caption = "Model 2: Pooled Logistic Regression (with Demographics)",
      booktabs = TRUE) %>%
  kable_styling(bootstrap_options = c("striped", "hover"),
                full_width = FALSE,
                latex_options = "HOLD_position") %>%
  row_spec(sig_rows, bold = TRUE) %>% # Bolds significant predictors like Pembrolizumab
  column_spec(1, italic = TRUE)

# 1. Identify agent columns
agents_to_join <- data %>%
  mutate(.id = as.integer(row_number())) %>%
  select(.id, starts_with("agent_"))

subset_long_data <- complete(imputed_data, action = "long", include = TRUE) %>%
  left_join(agents_to_join, by = ".id") %>%
  filter(dataset %in% c("ISPY2", "NACT")) %>%
  mutate(ethnicity = as.character(ethnicity),
         # Group Hispanic into 'Other' or 'Caucasian' (or whatever your baseline is)
         ethnicity = ifelse(ethnicity == "hispanic", "Other/Unknown", ethnicity),
         ethnicity = as.factor(ethnicity)) %>%
  select(pcr, .imp, .id, everything()) %>% # Bring pcr to the front to be sure
  mutate(.imp = as.integer(.imp), # as.mids often prefers integer .imp for indexing
         .id = as.integer(.id))

agent_cols <- names(subset_long_data)[grep("agent_", names(subset_long_data))]

# 2. Calculate Correlation Matrix
cor_matrix <- cor(subset_long_data[, agent_cols], use = "pairwise.complete.obs")

processed <- c()
collapsed_data <- subset_long_data

for (col in agent_cols) {
  if (col %in% processed) next

  # A. Identify Positive Collinearity (r > 0.99)
  identical_cols <- names(which(cor_matrix[col, ] > 0.99))

  # B. Identify Anti-Collinearity (r < -0.99)
  anti_cols <- names(which(cor_matrix[col, ] < -0.99))

  if (length(identical_cols) > 1) {
    # Combine Positive Collinearity
    new_name <- paste(identical_cols[1],
                      paste(gsub("agent_", "", identical_cols[-1]), collapse = "_"),
                      sep = "_")
    collapsed_data <- collapsed_data %>%
      mutate(!!new_name := get(col)) %>%
      select(-all_of(identical_cols))
    processed <- c(processed, identical_cols)

  } else if (length(anti_cols) > 0) {
```

```r
    # Combine Anti-Collinearity (Mutually Exclusive)
    # Name it: agent_DrugA_vs_DrugB
    new_name <- paste0(col, "_vs_", gsub("agent_", "", anti_cols[1]))

    collapsed_data <- collapsed_data %>%
      mutate(!!new_name := get(col)) %>% # 1 = DrugA, 0 = DrugB
      select(-all_of(c(col, anti_cols)))

    processed <- c(processed, col, anti_cols)
  }
}

# 3. Re-run MICE object creation
imputed_collapsed <- as.mids(collapsed_data)

# 4. Update your formula and fit
new_agent_vars <- names(collapsed_data)[grep("agent_", names(collapsed_data))]
final_agent_vars <- new_agent_vars[!new_agent_vars %in% c("agent_fec100")]
formula_str <- paste("pcr ~ bilateral_breast_cancer + multifocal_cancer + her2 +
                     tumor_subtype + age + bmi_group + menopause + ethnicity +",
                     paste(final_agent_vars, collapse = " + "))

fit3 <- with(imputed_collapsed, glm(as.formula(formula_str), family = binomial))
pooled_results3 <- pool(fit3)

reg_table_data <- summary(pooled_results3) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(Variable = term, P_Value = p.value)

# 2. Identify rows where p-value < 0.05
sig_rows <- which(reg_table_data$P_Value < 0.05)

# 3. Create the table
reg_table_data %>%
  kbl(caption = "Model 3: Pooled Logistic Regression (Clinical + Treatment Agents)",
      booktabs = TRUE) %>%
  kable_styling(bootstrap_options = c("striped", "hover"),
                full_width = FALSE,
                latex_options = "HOLD_position") %>%
  row_spec(sig_rows, bold = TRUE) %>% # Bolds significant predictors like Pembrolizumab
  column_spec(1, italic = TRUE)

# 1. Get predicted probabilities from each of the 5 models
# This creates a matrix of probabilities (one column per imputation)
predictions <- list()
for(i in 1:imputed_data$m) {
  # Extract the i-th completed dataset
  dat <- complete(imputed_data, i)

  # Fit the model on that specific dataset
  m_fit <- glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                 bmi_group + her2 + tumor_subtype + menopause,
               data = dat, family = binomial)
```

```r
  # Store the predicted probabilities
  predictions[[i]] <- predict(m_fit, type = "response")
}

# 2. Average the probabilities across the 5 imputations
avg_probs_m1 <- rowMeans(as.data.frame(predictions))

# 3. Calculate AUC using the actual 'pcr' values from any completed dataset
# (pcr is the same in all of them since it wasn't imputed)
actual_pcr <- complete(imputed_data, 1)$pcr
roc_curve <- roc(actual_pcr, avg_probs_m1)
m1_auc <- pROC::auc(roc_curve)

# 4. Results
par(pty = "s")

# 2. Plot the ROC curve with your specific requirements
plot(roc_curve,
     col = "brown1",
     lwd = 3,                        # Make the line thicker for readability
     main = paste("Pooled AUC-ROC for Model 1:", round(m1_auc, 3)),

     # Axis Labels
     xlab = "1-FPR",                 # Your requested X-axis label
     ylab = "TPR",                   # Your requested Y-axis label

     # Bounds and Scaling
     xlim = c(1, 0),                 # Standard ROC X-axis is 1 to 0 (FPR decreases)
     ylim = c(0, 1),                 # Y-axis bounds

     # Formatting
     legacy.axes = TRUE,             # Ensures the X-axis goes from 0 to 1 if preferred
     asp = 1,                        # Ensures an aspect ratio of 1 (strictly square)
     grid = TRUE                     # Adds a grid for easier value reading
)

# 3. Reset the plot parameters to default for future plots
par(pty = "m")

# 1. Get predicted probabilities from each of the 5 models
# This creates a matrix of probabilities (one column per imputation)
predictions <- list()
for(i in 1:imputed_data$m) {
  # Extract the i-th completed dataset
  dat <- complete(imputed_data, i)

  # Fit the model on that specific dataset
  m_fit <- glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                 bmi_group + age + her2 + tumor_subtype +
                 menopause + ethnicity,
               data = dat, family = binomial)

  # Store the predicted probabilities
  predictions[[i]] <- predict(m_fit, type = "response")
```

```r
}

# 2. Average the probabilities across the 5 imputations
avg_probs_m2 <- rowMeans(as.data.frame(predictions))

# 3. Calculate AUC using the actual 'pcr' values from any completed dataset
# (pcr is the same in all of them since it wasn't imputed)
actual_pcr <- complete(imputed_data, 1)$pcr
roc_curve <- roc(actual_pcr, avg_probs_m2)
m2_auc <- pROC::auc(roc_curve)

# 4. Results
par(pty = "s")

plot(roc_curve,
     col = "royalblue",
     lwd = 3,                        # Make the line thicker for readability
     main = paste("Pooled AUC-ROC for Model 2:", round(m2_auc, 3)),

     # Axis Labels
     xlab = "1-FPR",                 # Your requested X-axis label
     ylab = "TPR",                   # Your requested Y-axis label

     # Bounds and Scaling
     xlim = c(1, 0),                 # Standard ROC X-axis is 1 to 0 (FPR decreases)
     ylim = c(0, 1),                 # Y-axis bounds

     # Formatting
     legacy.axes = TRUE,             # Ensures the X-axis goes from 0 to 1 if preferred
     asp = 1,                        # Ensures an aspect ratio of 1 (strictly square)
     grid = TRUE                     # Adds a grid for easier value reading
)

par(pty = "m")

predictions <- list()

for(i in 1:imputed_collapsed$m) {
  # Extract the i-th completed dataset (which already has agents and filtering)
  dat <- complete(imputed_collapsed, i)

  # Fit the treatment model (Model 3)
  # We use the explicit variable list to ensure it matches your pooled summary
  m_fit <- glm(pcr ~ bilateral_breast_cancer + multifocal_cancer +
                 her2 + tumor_subtype + age + bmi_group +
                 menopause + ethnicity +
                 agent_paclitaxel + agent_ganitumab + agent_ganetespib +
                 agent_mk_2206 + agent_pembrolizumab + agent_amg_386 +
                 agent_trastuzumab + agent_neratinib + agent_abt_888_carboplatin +
                 agent_pertuzumab + agent_taxane,
               data = dat, family = binomial)

  # Store the predicted probabilities
  predictions[[i]] <- predict(m_fit, type = "response")
```

```r
}

# 2. Average the probabilities across the 5 imputations
avg_probs_m3 <- rowMeans(as.data.frame(predictions))

# 3. Calculate AUC using the actual 'pcr' values from the subset
# We pull pcr from the first imputation of the SUBSET
actual_pcr <- complete(imputed_collapsed, 1)$pcr
roc_curve_m3 <- roc(actual_pcr, avg_probs_m3)
m3_auc <- pROC::auc(roc_curve_m3)

# 4. Results
par(pty = "s")

plot(roc_curve_m3,
     col = "darkgreen",
     lwd = 3,                        # Make the line thicker for readability
     main = paste("Pooled AUC-ROC for Model 3:", round(m3_auc, 3)),

     # Axis Labels
     xlab = "1-FPR",                 # Your requested X-axis label
     ylab = "TPR",                   # Your requested Y-axis label

     # Bounds and Scaling
     xlim = c(1, 0),                 # Standard ROC X-axis is 1 to 0 (FPR decreases)
     ylim = c(0, 1),                 # Y-axis bounds

     # Formatting
     legacy.axes = TRUE,             # Ensures the X-axis goes from 0 to 1 if preferred
     asp = 1,                        # Ensures an aspect ratio of 1 (strictly square)
     grid = TRUE                     # Adds a grid for easier value reading
)

# 3. Reset the plot parameters to default for future plots
par(pty = "m")

# --- STAGE 1: Identify the clinical Subset IDs ---

# Extract the unique patient IDs from the ISPY2/NACT treatment subset
# We pull from the first imputation since patient_id is consistent across all
target_patient_ids <- complete(imputed_collapsed, 1)$patient_id

# --- STAGE 2: Refit Models 1 & 2 using the patient_id filter ---

# We refit on the full imputed_data object, but restrict it to the treatment IDs.
# Note: 'with' automatically handles the looping through all 10 imputations.
fit1_refit <- with(imputed_data,
                   glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                       bmi_group + her2 + tumor_subtype + menopause,
                       subset = patient_id %in% target_patient_ids,
                       family = binomial))

fit2_refit <- with(imputed_data,
                   glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
```

```r
                        bmi_group + her2 + tumor_subtype + menopause + age +
                        ethnicity,
                        subset = patient_id %in% target_patient_ids,
                        family = binomial))

# --- STAGE 3: Calculate the Valid AIC Comparison ---

# We compare these refitted models to your Model 3 (fit3)
aic_comparison <- data.frame(
  Model = c("Model 1 (Baseline)", "Model 2 (Demographic)", "Model 3 (Treatment)"),
  AIC = c(
    mean(sapply(fit1_refit$analyses, AIC)),
    mean(sapply(fit2_refit$analyses, AIC)),
    mean(sapply(fit3$analyses, AIC))
  )
)

# --- STAGE 4: Plot the Standardized Comparison ---
ggplot(aic_comparison, aes(x = Model, y = AIC, fill = Model)) +
  geom_bar(stat = "identity", width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = round(AIC, 1)), vjust = -0.5, fontface = "bold") +
  theme_minimal() +
  labs(title = "Model Selection: Standardized AIC Comparison",
       subtitle = paste0("N = ", length(target_patient_ids), " patients (ISPY2/NACT Subset)"),
       y = "Mean AIC",
       x = "") +
  scale_color_manual(values = c("brown1", "royalblue", "darkgreen")) +
  scale_fill_brewer(palette = "Set1")
# 1. Helper function to safely extract averaged probabilities from a 'mira' object
get_avg_probs <- function(mira_fit) {
  # Get the number of imputations (m)
  m_count <- length(mira_fit$analyses)

  # Extract predictions from each of the m models
  pred_list <- lapply(mira_fit$analyses, function(x) predict(x, type = "response"))

  # Average them across rows
  rowMeans(as.data.frame(pred_list))
}

# 2. Calculate probabilities for the two baseline models (Refit on Subset)
avg_probs_m1 <- get_avg_probs(fit1_refit)
avg_probs_m2 <- get_avg_probs(fit2_refit)

# 3. Get the actual pCR values from the subset
actual_pcr <- complete(imputed_collapsed, 1)$pcr

# 4. Create the PR multi-plot object
precrec_obj <- evalmod(
  scores = list(avg_probs_m1, avg_probs_m2, avg_probs_m3),
  labels = list(actual_pcr, actual_pcr, actual_pcr),
  modnames = c("Model 1: Baseline", "Model 2: Demographic", "Model 3: Treatment")
)
```

```r
# 5. Plot the Comparison
autoplot(precrec_obj, curvetype = "PRC") +
  theme_minimal() +
  scale_color_manual(values = c("brown1", "royalblue", "darkgreen")) +
  labs(title = "Precision-Recall Comparison: Clinical Impact",
       subtitle = "Evaluation of added predictive value on standardized subset",
       x = "Recall (Sensitivity)",
       y = "Precision (PPV)")
# 1. Get the pooled results into a standard dataframe
pooled_df <- as.data.frame(summary(pool(fit3), conf.int = TRUE))

# 2. Rename columns manually if they aren't standard
# Depending on your mice version, column names might be 'estimate' or 'est'
names(pooled_df)[names(pooled_df) == "estimate"] <- "est"
names(pooled_df)[names(pooled_df) == "2.5 %"] <- "low"
names(pooled_df)[names(pooled_df) == "97.5 %"] <- "high"

# 3. Exclude intercept and transform to Odds Ratios
plot_data <- pooled_df[pooled_df$term != "(Intercept)", ]
plot_data$OR <- exp(plot_data$est)
plot_data$OR_low <- exp(plot_data$low)
plot_data$OR_high <- exp(plot_data$high)

# 4. Generate the Forest Plot using standard ggplot2
ggplot(plot_data, aes(x = reorder(term, OR), y = OR)) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "steelblue") +
  geom_pointrange(aes(ymin = OR_low, ymax = OR_high), color = "darkgreen") +
  coord_flip() +
  theme_bw() + # theme_bw is very stable in older ggplot2 versions
  labs(title = "Model 3: Odds Ratios for pCR",
       x = "Predictor",
       y = "Odds Ratio (95% CI)") +
  scale_y_log10()
```